Ekdant

Selective SSM Storage with a Parallel Sandbox

for Memory-Efficient Reasoning

Three-stage text model: a selective state-space store (L1), a windowed dense stack (T), and a parallel debate sandbox with anchored gated descent (L2).

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Every numeric claim is either a configuration value, a derived parameter count, a hand-worked derivation, or marked `[PENDING]`.

Abstract

Transformer decoders scale quadratically in context length and their key–value (KV) cache grows without bound, so a model intended to reason over long inputs pays the most exactly where it should be strongest. Separately, a monolithic stack cannot say which component is responsible for a gain, and self-reported confidence is uncalibrated: agreement between reasoning paths is routinely mistaken for certainty even though agreement is trivially maximized by collapsing all paths onto one answer.

Ekdant is a three-stage text model that makes each of these addressable and, crucially, makes its own design falsifiable. Stage L1 is a selective state-space layer with three L1 state stores (content, control, sandbox), giving linear-time training and a fixed-size inference state independent of sequence length. Stage T is a plain dense Transformer stack—windowed grouped-query attention with rotary positions and SwiGLU feed-forward blocks—whose only job is to bound the KV cache and provide local exact retrieval. Stage L2 is a sandbox: K parallel trajectories that cross-read for R debate rounds and then perform S steps of *anchored gated descent* on an explicit energy before a single dense head produces logits. A consensus signal is written back into the L1 sandbox store, closing a memory/reasoning loop. L1 and L2 are separate stages; the store is SSM memory, the sandbox is the debate layer.

Two training signals are used, and the second one is the subject of a correction. We originally specified an agreement penalty between trajectories; we proved it is minimized by the very collapse it is meant to detect, and replaced it with a dimensionless collapse *floor* that is exactly zero while the hypotheses remain diverse. We also proved that the natural energy update—descending toward the running mean of the trajectories—is *readout-invariant* and therefore cannot change the model's output at all, and replaced it with an externally anchored, per-trajectory gated step that provably decreases each trajectory's energy and does move the readout. Both defects, their proofs, and their numerical consequences are reported rather than removed. Sizes are XS/S/M $\approx$ 12.4/21.0/46.0M parameters, trained from scratch on a single GPU. Every falsifiable claim in the supporting vault is backed by one of 66 locally stored papers or by an explicit hand derivation, and an explicit prior-art audit states which claims are *not* novel—including a compute-matched study in our own library that reports no reliable gain from the closest published analogue of stage L2.

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1 Introduction

1.1 The problem

Three failure modes motivate this work. They are stated separately because each has a different, non-interchangeable fix.

P1: Memory grows with the sequence. Self-attention compares every position with every other position, giving quadratic training cost in sequence length, and inference requires a KV cache that grows linearly in context length. A model meant to reason over long inputs—source codebases, medical records, papers with cross-references—is most expensive precisely on its intended inputs.

P2: A monolithic stack cannot be audited. When every component lives in one homogeneous stack, a benchmark gain cannot be attributed to a specific mechanism. Adding capacity and adding a mechanism are indistinguishable from the outside, so the resulting claim is unfalsifiable.

P3: Agreement is not confidence. A single autoregressive pass emits one answer with one probability distribution. When it is wrong, nothing records which step went wrong. A common remedy is to sample several reasoning paths and treat their agreement as a confidence signal. This remedy has a degenerate optimum that is rarely discussed: if all paths are made to agree, agreement is perfect and the model is confidently wrong. The fix is not to avoid agreement but to make the mechanism that *produces* agreement incapable of benefiting from collapse.

1.2 Contributions

1. **A minimal three-stage split** (L1 selective SSM storage $\rightarrow$ T dense local attention $\rightarrow$ L2 parallel sandbox) in which each stage exists to solve one of P1–P3, and each can be ablated independently at matched parameters and matched compute.

2. **Anchored gated descent (AGD)** for stage L2, introduced to replace a peer-mean energy update that we prove is readout-invariant. AGD admits a monotone-energy proof (Proposition 1) and is verified numerically.

3. **A dimensionless collapse floor** replacing a degenerate agreement penalty, with a proof that the original term is minimized at zero spread (Proposition 3) and a proof that the replacement is non-degenerate (Proposition 4).

4. **An audit that reports its own defects.** Four defects were found by internal review, each with a proof, a numerical demonstration, and a retained ablation that exhibits the broken behavior (Section 4).

5. **A self-limiting novelty claim.** A prior-art audit over 30 sandbox papers and 36 background papers concludes that *every ingredient* of stage L2 has precedent, that only the combination plus the memory write-back is unclaimed, and—importantly—that the closest compute-matched published study reports *no reliable gain* for its own version of this mechanism (Section 8).

6. **A falsifiability apparatus.** 66 locally stored papers, a claim-to-evidence index, hand-worked numerical examples requiring only a calculator, and a cost-matched evaluation protocol that fixes the comparison before any run is performed.

1.3 Research questions

RQ1 (memory) Does a fixed-size selective SSM state replace the KV cache while preserving long-range recall, and what does it cost in exact retrieval? Predicted outcome: recall is bounded, and we report the bound rather than hide it.

RQ2 (reasoning) At matched parameters and matched block-pass budget, do K debating trajectories plus consensus beat (a) a single path, and (b) a depth-matched model with no sandbox at all? The second control is not optional; see Section 8.

RQ3 (honesty) Is trajectory agreement calibrated against per-example accuracy, and can collapse be detected early by a monitored statistic rather than discovered post hoc?

1.4 Non-goals and scope

Ekdant is a text-only pretrained model studied at 12–46M parameters. It is not a tool-using agent (the ReAct line of work [38] is cited to draw this boundary explicitly), not multimodal, and not a mixture-of-experts model. At this scale, mixture-of-experts and sparse/dense hybrids add routing machinery without the capacity to exploit it; stage L2 spends the same parameters on hypothesis diversity instead. Every component is implemented in plain PyTorch with no external CUDA kernels, which keeps the thesis reproducible and auditable at the cost of raw speed (Section 9).

1.5 Notation

Symbol	Meaning
B	batch size
T	sequence length (2048 at train time, 4096 window)
V	vocabulary size (32768)
H	model width (256 / 320 / 512 for XS / S / M)
d	per-store state dimension d_{state} (96 / 128 / 256)
N	number of L1 blocks (2 / 4 / 4)
K	number of sandbox trajectories (4)
R	number of debate rounds (2)
S	number of descent steps (8)
W	attention window (512)
ρ	*spread ratio*, the dimensionless collapse monitor
$\rho_{\min}$	collapse floor threshold (0.05)

One naming caution, because two separate stages share a word. Stage L2 is *the sandbox*: the parallel debate layer. S^{sandbox} is one of the three state stores *inside* stage L1—a fixed-size SSM memory, not a layer. L1 and L2 are distinct stages, and their only coupling is the M3 write-back (§3.5), which writes L2's consensus into the S^{sandbox} store. Elsewhere in this document "sandbox" alone always means the L2 layer, and the store is always written with its $S^{\cdot}$ decoration or as "the L1 sandbox store".

2 Background

2.1 Selective state-space models

A classical linear state-space model maps an input signal through a latent state,

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad y(t) = Cx(t), \tag{1}$$

with no skip term ($D = 0$). Such a model is inherently recurrent and has constant memory, but for many years was empirically weak at long-range recall. Four results turned it into a competitive neural memory, and stage L1 follows them in order.

Structured parameterization and principled initialization. HiPPO theory [1] derives the optimal online projection of history onto polynomial bases, and S4 [3] builds a deep sequence model on that basis which handles 16K-length tasks.

Diagonal and simplified variants retain the power. Diagonal state matrices [4], multi-input multi-output formulations with a single scan [5], and S5 [6] match S4 with simpler implementations and parallelizable recurrences.

Input-dependent selection. Making the dynamics—the step size and the input/output projections—functions of the input allows the state to forget selectively. This is the step that closes copying and induction-style retrieval gaps and extrapolates far beyond training length [7]. It is the core mechanism of stage L1. The duality framework unifying this with attention, and the multi-head state-space design that makes the projections cheap, are in [8].

Hybridization as the honest fallback. Pure SSMs trail attention on exact associative recall, and hybrid models win there [18]. We treat this as a bound on our claims rather than a problem to engineer away: stage T exists partly to reintroduce local exact retrieval.
The wider family was surveyed and positioned rather than ignored: long convolutions [11], gated SSMs [12], retention [13], modern RNNs [14, 15, 16], byte-level selectivity [17], multidimensional states [9], and liquid time-constants [10]. The efficiency bar we hold ourselves to is IO-aware exact attention [19], and the standard benchmark framing for this family is the Long Range Arena [2].

2.2 The dense Transformer block

Stage T is the standard pre-norm decoder block of Vaswani et al. [21], with four well-understood modifications, each chosen for a specific reason.

RMSNorm and pre-norm. Root-mean-square (RMS) normalization [26] removes the mean-centering step of LayerNorm, and placing the norm before the residual [27] stabilizes deep stacks without a warmup-schedule hack. Both are used in every block.

Rotary position embeddings. Positions are injected by rotating query and key pairs by an angle that depends only on relative distance [22]. Because the rotation composes, a position m apart is represented by a rotation of $m\theta$, which is what makes extrapolation and window-relativity behave. No learned absolute position table is used, so there is no table to grow with context. For the same reason RoPE frequency rescaling is left *off* by default: with a 4096 window and stage-T depths of $T \leq 2048$, plain RoPE is already interpolating, so the extension factors of YaRN [29] would be tuned constants with no failure to repair. The mechanism stays available in the configuration for longer-context runs, and the factors actually used are recorded in the reproducibility appendix.

Grouped-query attention. Multi-query attention [23] shares a single key/value head across all query heads; grouped-query attention [24] interpolates between that and full multi-head attention by using a small group. We use 8 query heads and 2 key/value heads. The KV cache therefore costs $2 \times T \times$ head entries per layer instead of $8 \times T \times$ head, a $4\times$ reduction in the dominant cache term at unchanged attention quality in the cited results.

SwiGLU feed-forward. The feed-forward block uses a gated linear unit with a SiLU activation [25]: two parallel projections into an intermediate space, elementwise product, then projection out. This parameterization consistently outperforms a plain ReLU MLP at fixed parameter count.

Windowed attention. Attention is restricted to a window of $W = 512$ preceding tokens [28]. This is what actually bounds the cache: cost per layer is $O(TW)$ rather than $O(T^2)$, and memory is $O(W)$ per sequence rather than $O(T)$. The exact bound holds when the softmax is computed exactly; an IO-aware fused kernel reduces memory traffic without changing the math [20], which is why we specify the mathematics rather than a particular attention kernel. Sliding-window attention is also the natural partner of recurrent state, since the two cover complementary distance ranges.

2.3 Multi-hypothesis reasoning and consensus

Reasoning traces improve prediction [30]; sampling several and voting beats one greedy path at matched budget [31]; letting paths read and revise each other helps further [32, 33]; keeping live alternatives with aggregation beats resampling only at the end [35, 36]; and decomposing hard problems into easier ones needs no planner [37]. Reflection loops can be folded into latent refinement [39, 40], and per-step supervision beats outcome-only supervision for reranking [41, 42], with self-generated rationales usable as targets [43, 44]. Multi-agent debate has also been cast explicitly as a evaluation of communication rather than of a single reasoner [45], which is the same move stage L2 makes: the object of study is whether independently generated trajectories inform each other, not whether any one of them is correct.

Two constraints from this literature shape the evaluation rather than the architecture. First, cost honesty is load-bearing: debate systems frequently *tie* voting ensembles at equal budget and are sensitive to tuning [47], so every result in Section 7 is reported cost-matched with tokens and FLOPs stated, never against greedy decoding alone. Second, calibration is the deciding measurement [64]: agreement is only a useful confidence signal if it predicts per-example accuracy, which is exactly RQ3.

2.4 Energy-based refinement

Refinement can be framed as descending an energy to a fixed point rather than as a stack of residual layers. The Energy Transformer iterates a block to reduce a global energy [54], and recent work performs a fixed number of gradient steps on a *parameterized* energy as the forward pass [55]. Latent-thought calibration pushes intermediate representations toward low-energy, high-consistency regions [48]. This is the direct ancestor of stage L2's descent, and Section 4.1 records a failure mode we hit when implementing it naively.

2.5 Collapse in reinforcement learning, and why agreement penalties are degenerate

This subsection exists because it determines a design decision in Section 3.6. In maximum-entropy reinforcement learning, the standard regularizer trades reward against divergence from the reference policy, and there is a large literature documenting that this trade *encourages* mode collapse: the KL term's pressure toward the reference can be satisfied by reducing the number of distinct solutions rather than by improving any of them [57], and entropy collapse is documented in RL-with-verifiable-rewards pipelines [58].

The mechanism transfers directly to our setting. If a training term rewards agreement between trajectories, the cheapest way to reduce it is to make all trajectories identical—which destroys the diversity the term was introduced to exploit. This is not a tuning problem; it is an optimization

direction pointing at the wrong optimum. Proposition 3 in Section 4.2 states and proves the point for our exact loss, and Section 3.6 gives the replacement.

2.6 What is claimed as novel, and what is not

The prior-art audit (`04-Model-DeepDive/_Prior_Art_and_Novelty_Check.md`) reached an uncomfortable but clear conclusion: every ingredient of stage L2 has published precedent.

Ingredient	Precedent
Parallel hypothesis streams in one model	Multi-Stream Transformers [49]
Streams that read and revise each other	Talking-Heads Attention [50]
Function-separated streams at small scale	Dual-Stream Transformer [51]
Iterative latent refinement, weighted readout	Autoregressive TRM [52]; Coconut; original TRM [53]
Descending an energy to a fixed point	Energy Transformer [54]; NRGPT [55]
Layered proposal/aggregation	Mixture-of-Agents [56]
Divergent initialization for multiple paths	Divergent Debate [33]
Rationale-aware early stopping	RASC [46]

Accordingly the novelty claim is restricted to the *combination* and to one specific mechanism: multi-trajectory tensor-level debate placed after a dense backbone, whose consensus is written back into an SSM state store. No single component is claimed as new. Section 8 discusses the strongest published evidence against even this.

3 Method

3.1 Overview

The forward pass is a single chain with no external calls:

$$\begin{aligned} \texttt{input_ids} &\rightarrow \text{L1 (SSM storage)} \rightarrow \text{T (dense local attention)} \\ &\rightarrow \text{L2 (sandbox)} \rightarrow \text{output head.} \end{aligned} \tag{2}$$

Shapes: L1 and T each map $[B, T, H] \rightarrow [B, T, H]$; L2 maps $[B, T, H] \rightarrow [B, T, H]$ through K trajectories of the same shape. No hidden states grow with T at inference time.

Figure 2 draws that chain, annotates what is bounded and what is not, and shows the one non-obvious edge: the M3 write-back that returns L2's consensus into L1's third state store. That dashed edge is what makes this a memory/reasoning loop rather than a three-block sandwich.

Every diagram in this document uses one fixed colour code, given once here and never varied: *blue* is L1 and therefore storage, *green* is stage T and therefore local dense compute, *orange* is L2 and therefore reasoning, *red dashed* marks the feedback path, and *grey* is a neutral tensor or projection that belongs to no stage. The same code colours the row groups of the tables, so a reader can tell at a glance which stage a number belongs to. Figure 1 is the key.

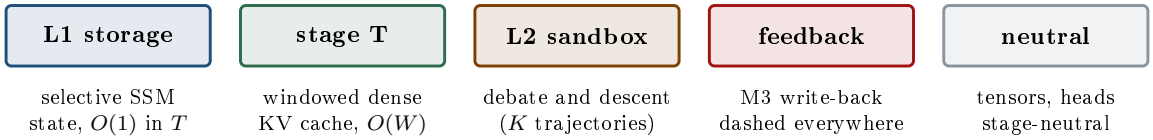


Figure 1: The colour code used in every figure and in the row groups of every table. Colour is redundant with position and with the labels themselves—it is a reading aid, never the sole carrier of meaning—but it is applied consistently from this point on.

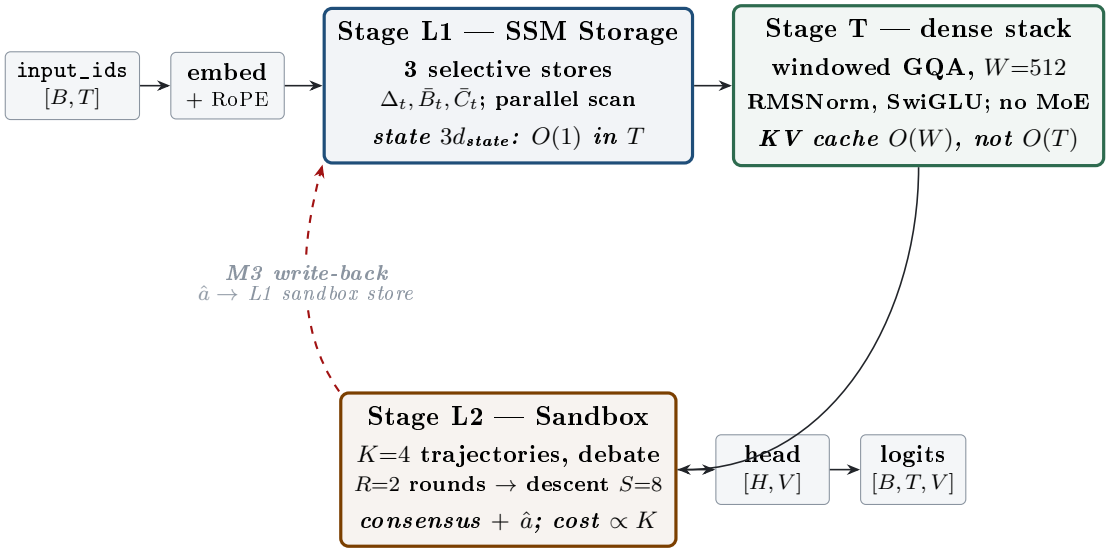


Figure 2: The three-stage forward pass. Solid arrows are the forward chain; the dashed red edge is the M3 write-back, which carries L2's consensus back into L1's sandbox store, so that reasoning and memory are coupled rather than merely adjacent. The italic line inside each stage states that stage's memory cost, which is the whole justification for the layout: L1 contributes a constant-size state, stage T caps the cache at a fixed window, and L2 adds work proportional to K but never to T.

3.2 Layer 1: selective SSM storage

3.2.1 Discretization, and an honest note about conventions

Discretizing (1) with a zero-order hold over a step Δ gives $\bar{A} = \exp(\Delta A)$ and

$$\bar{B} = (\Delta A)^{-1}\big(\exp(\Delta A) - I\big)\,\Delta B, \tag{3}$$

which for a single diagonal entry reduces to $\bar{B} = \big((e^{\Delta A} - 1)/A\big)B$.
We instead implement the first-order form $\bar{B} = \Delta B$, which is the convention in the Mamba reference recurrence [7]. Expanding (3),

$$\frac{e^{\Delta A} - 1}{A} = \Delta\Big(1 + \tfrac{\Delta A}{2} + \tfrac{(\Delta A)^2}{6} + \cdots\Big), \qquad \text{so} \qquad \bar{B} = \Delta B + O(\Delta^2). \tag{4}$$

This is not cosmetic, and we state the magnitude rather than assume it away. On a three-step hand check with $A = -1$ and Δ reaching 1.0, the first-order form inflates the final state by **43%** (1.1693 versus the exact 0.8173; both computed by hand in `28_Worked_Examples.md`). At our initialization range $\Delta \in [0.001, 0.1]$ the relative error is $\approx \Delta/2 < 0.5\%$ and therefore harmless. Two rules follow, and are enforced throughout this document: never quote exact-ZOH numbers from a ΔB implementation or vice versa, and record Δ alongside any scan statistic.

3.2.2 Selection

The dynamics become functions of the input. Per position t and channel c:

$$\Delta_t^c = \mathrm{softplus}\big(W_{\Delta,c}\,u_t + b_{\Delta,c}\big) + \epsilon, \tag{5}$$
$$\bar{B}_t^c = \Delta_t^c\, B^c(u_t), \tag{6}$$
$$\bar{A}_t^c = \exp\big(\Delta_t^c A_c\big), \qquad A_c = -\,\mathrm{softplus}(A_{\log,c}), \tag{7}$$
$$x_t^c = \bar{A}_t^c \odot x_{t-1}^c + \bar{B}_t^c \odot u_t, \qquad y_t^c = C^c(u_t) \odot x_t^c. \tag{8}$$

The learned ϵ sets the initial step range; the term $\mathrm{softplus}(\cdot)$ guarantees $\Delta > 0$, which is required for $\bar{A} = \exp(\Delta A) \in (0, 1)$ and hence for forgetting rather than divergence. Because Δ, B, and C all depend on u_t, the amount of information retained per token is content-dependent: this is the selective mechanism of [7], and it is the only reason stage L1 is not simply a fixed linear recurrence.

3.2.3 Three stores

A single shared trunk $u_t = \mathrm{DWConv}(\mathrm{SiLU}(\mathrm{Linear}(h_t)))$ feeds *three* selective channels, all inside stage L1, scanned in parallel. They are given different step-size biases so that they specialize, and each keeps its own A, Δ, B, C, recurrence, readout, gate, and norm. The third store is named S^{sandbox} only because it tracks stage L2; it is an L1 state store, not the L2 layer.

Store	Step regime	Function
S^{content}	slow Δ	retains *what was said*; long memory over discourse
S^{control}	fast Δ	retains *what to do*; instruction and task features that must update quickly
S^{sandbox}	medium Δ	tracks stage L2: scans distilled task features plus the L2 write-back f

Multi-channel states have precedent in S5 [6] and Mamba-2's multi-head design [8]; the functional specialization of the channels is ours. Figure 3 draws one step of the block: the same input

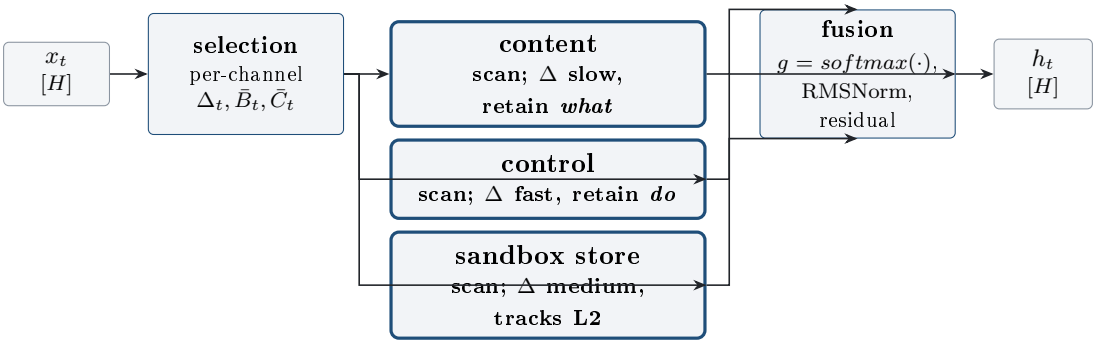


Figure 3: One step of the L1 block. A single input x_t feeds a selection network that emits per-channel step sizes; the three stores then integrate the same token at deliberately different rates, which is what makes the layer *selective* rather than a fixed-rate recurrence. Each store is advanced by its own parallel scan, the three outputs are combined by a learned gate and a residual connection (Eq. (9)), and the result is a fixed-size state plus a h_t that continues to stage T.

produces three states at three different forgetting rates, and the fusion gate decides how much of each reaches the residual stream.
The outputs are fused with learned per-channel weights and a residual connection:

$$h_t = \mathrm{RMSNorm}\Big(\sum_c g_t^c \odot y_t^c + h_t^{\mathrm{in}}\Big), \qquad g = softmax\big(\mathrm{Linear}_{\mathrm{fuse}}(h_t^{\mathrm{in}})\big), \tag{9}$$

with an output gate $y' = y \odot \mathrm{SiLU}(\mathrm{Linear}_{\mathrm{gate}}(x))$ applied per channel before fusion. We additionally emit a pooled *compressed* triple $\texttt{compressed} = [c_1; c_2; c_3] \in \mathbb{R}^{B\times 3d}$ for logging and for the bounded-memory verification of Section 7.3; this tensor is diagnostic only and never feeds forward.

3.2.4 Initialization

Following the S4D-Lin convention [4] we set $A_{\log,c} = \log(c)$ for channel $c = 1 \dots d$ and $A_c = -\mathrm{softplus}(A_{\log,c})$, which spreads the time constants geometrically so that different channels cover different memory horizons. The step bias is set so that $\Delta \in [0.001, 0.1]$ at initialization (see the error analysis above). Projections B and C use Xavier initialization, and gates are zero-initialized so that every channel starts open—contribution is added, not gated away, at step zero.

3.2.5 Parallel scan versus recurrence, and the $O(1)$ claim

Equations (7)–(8) define a recurrence, but training does not use it. Expanding the recurrence gives

$$x_t^c = \sum_{s\le t}\Big(\prod_{s<r\le t} \bar{A}_r^c\Big)\bar{B}_s^c \odot u_s \;+\; \Big(\prod_{r\le t}\bar{A}_r^c\Big) \odot x_0^c. \tag{10}$$

This is a first-order linear recurrence with input-dependent coefficients, so it is computable by an associative scan in $O(T \log T)$ work (or $O(T)$ with a sequential-but-batched formulation), fully parallel, with autograd flowing through every term. At generation time the same equation is evaluated by carrying the state: the model holds $x^c \in \mathbb{R}^{B\times d}$ per channel, a *fixed* tensor independent of how many tokens have been produced.
This yields the precise version of the "$O(1)$ inference" claim, which we state carefully because the loose version is wrong. The inference *state* is $O(1)$ in T. Total work and total attention memory are not: stage T still performs $O(TW)$ attention work and holds an $O(W)$ KV window per layer. What is eliminated is the *unbounded, per-token* cache and the quadratic training cost. *Remark:* the honest summary is "bounded state, linear training, windowed attention", not "constant cost".

3.3 Stage T: dense Transformer stack

N_T identical blocks ($N_T = 3/5/6$ for XS/S/M), each pre-norm:

$$\begin{aligned} u_1 &= \mathrm{RMSNorm}(x) + \mathrm{PE}, \quad & x &\leftarrow x + \mathrm{GQA}\big(u_1\big) \\ u_2 &= \mathrm{RMSNorm}(x), & x &\leftarrow x + \mathrm{SwiGLU}\big(u_2\big), \end{aligned} \tag{11}$$

with PE the rotary injection and GQA restricted to the window W. All blocks are identical: no dense/sparse split, no sparse indexer, no experts. The rationale is that at 12–46M parameters those mechanisms add machinery without the capacity to exploit it, and every ablation we can afford is better spent on the L2 question.

3.4 Layer 2: the sandbox

3.4.1 Diverse initialization

K trajectories are produced by K distinct projections,

$$T_0^{(k)} = W_k\, \mathtt{hT} + b_k, \qquad k = 1, \ldots, K, \tag{12}$$

with W_k mutually orthogonal and spread gains. Identical copies of `hT` would satisfy any agreement objective trivially while destroying the diversity that motivates multiple paths [33]; diverse initialization is therefore load-bearing, not cosmetic, and we ablate diverse-versus-copy initialization explicitly.

3.4.2 Debate rounds

For round $r = 1, \ldots, R$ and trajectory k, the trajectory reads the others—not itself:

$$Q_k = T^{(k)} W_Q, \qquad K_j = T^{(j)} W_K, \quad V_j = T^{(j)} W_V \quad (j \neq k)$$

$$\mathrm{read}_k = \mathit{softmax}\Big(Q_k K_{-k}^{\top} / \sqrt{\mathrm{head}}\Big) V_{-k} \tag{13}$$

$$\gamma_k = \sigma\big(W_\gamma [\mathrm{read}_k \,;\, T^{(k)}] + b_\gamma\big)$$

$$T^{(k)} \leftarrow (1 - \gamma_k) \odot T^{(k)} + \gamma_k \odot \mathrm{MLP}(\mathrm{read}_k). \tag{14}$$

The repair gate γ_k lets a trajectory decline the crowd, which is necessary for the debate to be able to correct rather than merely average. The cross-attention in (13) is over $K - 1 \leq 7$ keys, so it is negligible cost and needs no specialized kernel.

3.4.3 Anchored gated descent

This subsection is the core technical contribution, and it exists because the obvious design is provably inert.

The step. Define a token-conditioned anchor *outside* the trajectory set, an energy per trajectory, and a learned per-trajectory per-channel gate:

$$a_t = W_a\, \mathtt{hT}_t, \tag{15}$$

$$E_k = \tfrac{1}{2} \big\| T^{(k)} - a_t \big\|_2^2, \tag{16}$$

$$g_k = \sigma\big(W_g [\, T^{(k)}; a_t \,] + b_g\big) \in (0,1)^H, \tag{17}$$

$$T^{(k)} \leftarrow T^{(k)} - \eta\, g_k \odot \big(T^{(k)} - a_t\big), \qquad \eta \approx \tfrac{0.5}{S}. \tag{18}$$

Why this step is well posed.

Proposition 1 (Monotonicity of the gated step)**.** *Under* (18) *with* $0 \le \eta g_k \le 2$*, the energy* (16) *is non-increasing:* $E_k^{new} \le E_k$*, with strict decrease whenever* $\eta g_k \in (0,2)$ *and* $T^{(k)} \neq a_t$*.*

Proof. Write $c = \eta g_k \odot$ (elementwise) and $\delta = T^{(k)} - a_t$. Then $T^{(k)}_{\text{new}} - a_t = (1-c) \odot \delta$, so

$$E_k^{\text{new}} - E_k = \tfrac{1}{2}\big(\|(1-c)\delta\|^2 - \|\delta\|^2\big) = \tfrac{1}{2}\sum_d \big[(1-c_d)^2 - 1\big]\delta_d^2 = \sum_d \big(c_d^2 - 2c_d\big)\,\frac{\delta_d^2}{2}. \tag{19}$$

Each summand is ≤ 0 when $0 \le c_d \le 2$, and < 0 when $c_d \in (0,2)$ and $\delta_d \neq 0$. □

Remark 1 (What kind of descent this is). (18) is *gate-scaled (preconditioned) descent* on E_k, not literally $-\eta \nabla E_k$, and the two coincide only when $g_k \equiv 1$. We use the word "preconditioned" throughout and never claim exact gradient descent.

Why the readout moves. Two independent mechanisms guarantee the consensus changes. First, the target a_t is *external* to the trajectory set, so the update is not a redistribution among trajectories. Second, the gate g_k differs across k, so the update is not a uniform linear map applied identically to all trajectories. Either alone would suffice; together they make the consensus $\bar{T}^{(S)}$ a genuine function of the backbone state.

The same heterogeneity is also what preserves spread. This second role is easy to miss. Write $T^{(k)} = a_t + \lambda_k\big(T_0^{(k)} - a_t\big)$, where λ_k is the product of $1 - \eta g_k$ over the descent steps. Were every λ_k equal, the deviations from the consensus would scale by λ_k alone and `trajectory_variance` would decay as $(1-\eta)^{2S}$—precisely the 0.356 of Section 4.1, since $\eta = 0.5/S$. When the λ_k differ, however, one term survives and never contracts. With $d_k = T_0^{(k)} - \bar{T}_0$ and $u = \bar{T}_0 - a_t$, $T^{(k)} - \bar{T} = \lambda_k d_k - \bar{\lambda} d + (\lambda_k - \bar{\lambda})\,u$, and the last term is proportional to $\mathrm{Var}_k(\lambda_k)$ and does not decay at all. Heterogeneous gates therefore convert descent pressure into persistent spread instead of collapse. Measured, V_{pre} retains 0.356 under identical gates, 0.396 under a common gate $g \equiv 0.9$, and 0.60–0.75 under heterogeneous gates—the regime containing the healthy run, which falls $1.95 \to 1.22$, a factor of 0.626. Gate heterogeneity is thus load-bearing twice over: it is what makes the readout move, and it is what holds ρ clear of the floor.
Measured on the toy model of `28_Worked_Examples.md`: the consensus moved by $\|\Delta\bar{T}\|_2 = 0.2295$, closing 0.2199 of its distance to the anchor, with per-trajectory deviation retention [0.827, 0.822, 0.744, 0.787]—i.e. refinement that did not force collapse. The inert peer-mean variant moved the consensus by exactly 0.

3.4.4 Consensus and readout

The consensus is the mean of the S trajectories (the default), or alternatively the lowest-anchor-energy trajectory $\arg\min_k \|T^{(k)} - a_t\|$. A single dense projection then produces logits:

$$y = \bar{T}^{(S)} \in \mathbb{R}^{B\times T\times H}, \qquad \text{logits} = y\,W_{\text{vocab}} \in \mathbb{R}^{B\times T\times V}, \tag{20}$$

with W_{vocab} tied to the input embedding. The head is evaluated once per micro-batch.

Figure 4 shows the whole of stage L2. The two halves are deliberately asymmetric: the debate phase is *horizontal* (trajectories interact with each other and the anchor is not yet defined), whereas the descent phase is *vertical* (every trajectory moves toward the same external anchor a_t). The anchor is what makes the second half capable of changing the readout at all, which is the subject of Section 4.1.

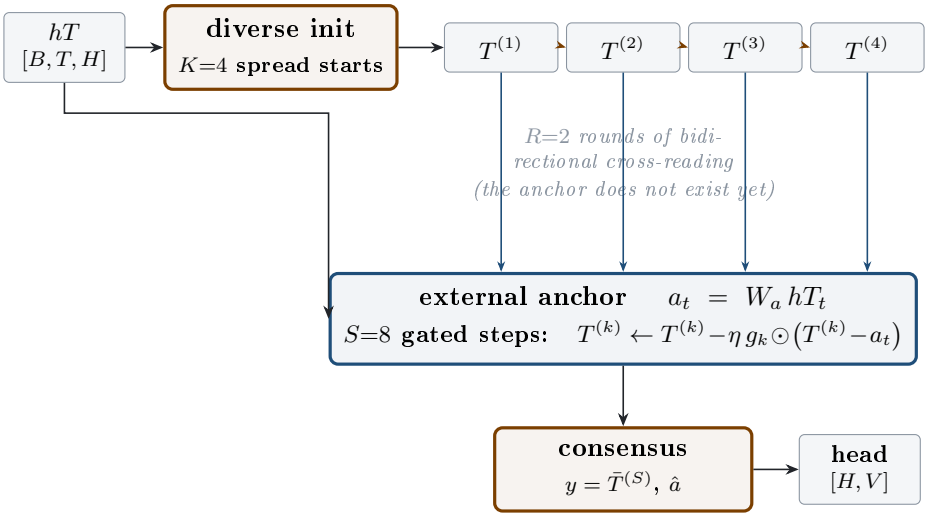


Figure 4: Stage L2 in full. Diverse initialization separates K trajectories; the debate phase lets them read and revise each other (R rounds, bidirectional arrows) while the anchor does not yet exist. Only then is $a_t = W_a hT_t$ computed, and all K trajectories descend toward that single external point for S steps under per-trajectory gates. The consensus and the agreement diagnostic $\hat{a}$ leave as the M3 write-back.

3.4.5 The three reported quantities

Three statistics ship with every forward pass. They are deliberately given distinct names because conflating them is what made an earlier version of this design undetectable.

`anchor_energy`, $E = \text{mean}_{k,t} E_k$. The quantity AGD descends. Reported as a monitor; *never* used as a loss term, because minimizing it directly collapses every trajectory onto the anchor.

`trajectory_variance`, V_{pre}. Mean squared distance of each trajectory to the consensus, measured *after* the debate rounds and *before* descent. This ordering matters: measuring after descent would report the effect of the descent rather than whether diversity survived cross-examination.

`spread_ratio`, ρ. Dimensionless: the ratio $\rho = V_{\text{pre}} / \text{mean}_{t,d}(\texttt{hT}^2_{t,d})$, i.e. the fraction of backbone activation energy that the K hypotheses still span. Because both numerator and denominator scale with activation magnitude, ρ is comparable across width H, length T, and training scale. Collapse is $\rho \to 0$.

3.5 The M3 write-back, and the causality rule

Stage L2 is not a terminal head. After the consensus is formed, it emits a feedback signal that is written into L1's third state store, S^{sandbox}:

$$f_t = \text{Linear}_{\text{fb}}([\, y_t \,;\, V_t \,]), \qquad x^3_t = \bar{A}^3_t \odot x^3_{t-1} + \bar{B}^3_t \odot u_t + F \odot f_{t-1}. \tag{21}$$

This closes the loop between reasoning and memory: the sandbox can carry tracking state and inject instructions into subsequent processing. Two constraints are enforced and are not negotiable:

1. **Causality.** Position t reads only states at positions $< t$. The per-position state m^3_t is the only thing stage L2 ever reads; the pooled `compressed` triple is diagnostic and never feeds forward.

2. **No cross-batch leakage.** During training, f arrives *detached* from the previous micro-batch. This is stated rather than hidden because the alternative silently conditions on future minibatch content.

3.6 Training objective

There are exactly two training signals:

$$\mathcal{L} \;=\; \lambda_{\text{lm}} \cdot \text{CE}\big(\text{shift}(\text{logits}),\, \text{shift}(\text{labels})\big) \;+\; \lambda_{\text{sbx}} \cdot \big[\max\big(0,\, \rho_{\min} - \rho\big)\big]^2, \tag{22}$$

with $\lambda_{\text{lm}} = 1.0$, $\lambda_{\text{sbx}} = 0.1$, $\rho_{\min} = 0.05$. Cross-entropy is causal and the head runs once per micro-batch.

The second term is a *floor*, not an agreement penalty. It is exactly zero while the hypotheses retain at least $\rho_{\min}$ of the backbone's spread, and grows only once they have collapsed. Verified numerically: a healthy run ($\rho : 1.95 \to 1.22$) pays 0; a collapsed run ($\rho = 10^{-6}$) pays $(0.05 - 10^{-6})^2 = 0.0025$.

Figure 5 plots the penalty. The shape is what justifies calling it a floor rather than a regularizer: the function is identically zero across four orders of magnitude of healthy spread, so it contributes no gradient at all during normal training, and only the extreme right tail—where the hypotheses have genuinely collapsed—is penalized. The penalty is also bounded, at $(\rho_{\min})^2 = 0.0025$, so even total collapse cannot dominate the cross-entropy term.

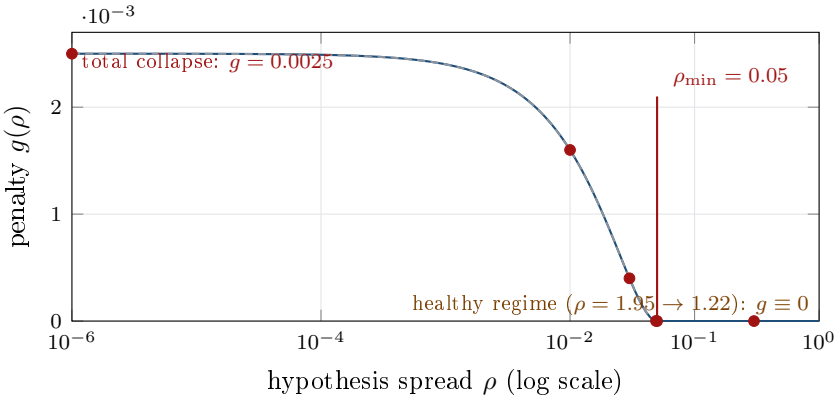


Figure 5: The spread floor $g(\rho) = [\max(0, \rho_{\min} - \rho)]^2$ on a logarithmic spread axis. Solid blue is the penalty; the dashed segment marks the region in which it is active. It is exactly zero for all $\rho \geq \rho_{\min}$, so it injects no gradient while the hypotheses remain distinct, and it saturates at 0.0025 under complete collapse—two and a half orders of magnitude below the $\lambda_{\text{lm}} = 1$ cross-entropy term, so it can flag a collapse without being able to cause one.

3.7 Complexity and parameter budget

Cost. L1 training is $O(T)$ per channel with the associative scan (10); stage T is $O(TW)$ attention plus $O(THI)$ feed-forward; L2 is

$$O\big(RKTH^2 \;+\; RK^2TH \;+\; SKTH\big), \tag{23}$$

comprising the K trajectory inits, the $3RK$ debate projections and the single anchor projection (the $O(TH^2)$ terms), the R rounds of K-way cross-read at $O(K^2TH)$, and the gates of the S descent steps at $O(SKTH)$. The projection terms dominate whenever $H \gg K$: at configuration M (H=512, K=4, S=8, R=2) they are 128× the cross-read term, so counting only the K^2 attention work understates L2 by more than two orders of magnitude. L2 is nonetheless linear in T and quadratic only in the tiny constant $K \leq 8$. Inference state is $3d$ per L1 channel plus an $O(W)$ KV window per T layer, independent of tokens generated.

That constant-plus-window claim is easier to believe as a curve than as an asymptotic, so Figure 6 plots the inference-side state footprint against sequence length at configuration M. The windowed curve is flat from $W = 512$ onward because W is fixed; the full-attention curve is the same curve with the window removed, and it is the only term that grows.

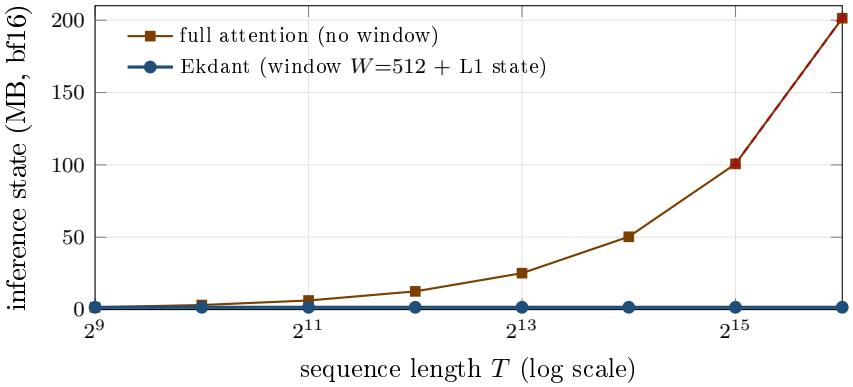


Figure 6: Analytic projection of the inference-side state footprint at configuration M (H=512, N=4 L1 layers, N_{T}=6, d_{state}=256, 2 KV heads, bf16). Ekdant holds flat at 1.58 MB: a 6 KB L1 recurrent state ($3 \cdot 256 \cdot 2$ bytes per L1 layer, over $N = 4$ layers) plus a KV window that saturates at $W = 512$. Full attention over the same tokens is 3 KB per token, growing linearly without bound—4.0 times larger at $T = 2048$ and 127 times larger at $T = 65536$. Weights, activations and the K-trajectory sandbox scratch space are excluded; the figure isolates the one term the architecture is designed to make constant.

Parameters. Embeddings are tied at $V \cdot H$. One T block costs approximately $2.5H^2$ (grouped-query projections) $+ 3HI$ (SwiGLU with $I = 2H$).

Component	XS (H=256)	S (H=320)	M (H=512)
Embeddings (tied)	8.4M	10.5M	16.8M
L1 aSSM (N=2/4/4)	1.5M	4.8M	13.0M
T (N_T=3/5/6)	1.7M	4.4M	13.4M
L2 (init + debate + anchor W_a + gate W_g)	0.8M	1.3M	2.8M
Output head	tied	tied	tied
Total	**12.4M**	**21.0M**	**46.0M**

All figures are closed-form estimates and must be confirmed with the `estimate_model_size` utility before any results table is published. The L2 figure rose from 0.6/1.0/2.2M when the peer-mean step was replaced by AGD, because the anchor $W_a \in \mathbb{R}^{H \times H}$ and gate $W_g \in \mathbb{R}^{2H \times H}$ are new parameters; the budget increase is the honest cost of the correction.

4 Design audit: four defects found and fixed

This section reports defects found in our own design during review. Each is given as a proposition with proof, a numerical demonstration, and a retained ablation. None is described as a limitation of the approach in general; all four are limitations of an earlier version of *this* approach, found before the main runs.

4.1 Defect 1: the energy loop was readout-invariant

The initial design descended each trajectory toward the running mean of the trajectories:

$$T^{(k)} \leftarrow (1 - \eta)\, T^{(k)} + \eta\, \bar{T}, \qquad \bar{T} = \frac{1}{K} \sum_{j} T^{(j)}. \tag{24}$$

This is a very natural thing to write, and it is useless.

Proposition 2 (Peer-mean descent is mean-preserving)**.** *Under* (24) *with common step* η *and* any *doubly stochastic mixing matrix* A*, that is* $y^{(k)} = \sum_j A_{kj} T^{(j)}$ *with* $\sum_j A_{kj} = 1$ *and* $\sum_k A_{kj} = 1$*, the mean* $\bar{T}$ *is invariant. In particular* $\bar{T}_{new} = \bar{T}$*, so any readout that depends only on* $\bar{T}$ *is unchanged.*

Proof. Average (24) over k with common η:

$$\bar{T}_{\text{new}} = (1-\eta)\bar{T} + \eta\bar{T} = \bar{T}. \tag{25}$$

More generally, for $y^{(k)} = \sum_j A_{kj} T^{(j)}$ with row sums 1, we get $\bar{T}_{\text{new}} = (1-\eta)\bar{T} + \eta\bar{T}$ by *double stochasticity*: averaging the targets gives $\frac{1}{K}\sum_k y^{(k)} = \sum_j \big(\frac{1}{K}\sum_k A_{kj}\big) T^{(j)} = \sum_j \frac{1}{K} T^{(j)} = \bar{T}$, using column sums 1. □

Remark 2 (The precise form, because the loose form is false). It is *not* true that any uniform pull toward any linear function of the trajectories is mean-preserving. For a general step $T^{(k)} \leftarrow T^{(k)} + \eta(y^{(k)} - T^{(k)})$ we get

$$\bar{T}_{\text{new}} = \bar{T} + \eta\big(\bar{y} - \bar{T}\big), \tag{26}$$

so the consensus moves *unless* $\bar{y} = \bar{T}$. Column sums equal to 1 is exactly the condition guaranteeing that, and leave-one-out mixing $(J-I)/(K-1)$ satisfies it—which is why every peer-mean variant collapses onto the same inert fixed point. A non-doubly-stochastic A gives $\bar{y} \neq \bar{T}$ and the readout *does* move. Design rule: the descent target must be external to the trajectory set, as a_t is in (15), or a non-linear function of it, as the gate is.

Consequences, measured. Because the head reads only the consensus (20), logits(S=0) = logits(S=8) *exactly*—gradient descent that provably cannot change the output. Meanwhile the spread decays like $(1-\eta)^{2S} = 0.356$ at the configured $\eta = \frac{0.5}{S} = 0.0625$, $S = 8$: the model became less diverse while producing bit-identical logits. A pure accuracy comparison at $S = 0$ versus $S = 8$ would have shown "no effect" and been interpreted as evidence that sandbox refinement does not help, when the truth is that the mechanism was disconnected from the output.

Figure 7 makes the asymmetry visible. Both panels matter, and the point is precisely that the obvious diagnostic fails. Energy—the quantity the refinement optimizes—falls in *both* variants, and the inert peer-mean variant actually falls *faster*. What distinguishes them is the readout: AGD moves the consensus toward the anchor, while peer-mean leaves it bit-identical forever, because averaging a fixed set of trajectories is a fixed point of the step. So "energy went down" would have certified the broken variant as the more successful one.

Fix and retained ablation. AGD (18) replaces (24); the anchors are external and the gates differ across k, so by (26) the consensus moves. The inert variant is *retained* as a reported contrast ablation rather than deleted, because it is the cleanest available demonstration that a refinement mechanism must be connected to the readout to be measurable.

4.2 Defect 2: the agreement loss was minimized by collapse

The initial second loss term was an agreement penalty λC, where C was the mean squared deviation of trajectories from their consensus.

Proposition 3 (The agreement penalty is minimized at zero spread)**.** *Let* $T^{(1)}, \ldots, T^{(K)} \in \mathbb{R}^H$ *and* $C = \frac{1}{K}\sum_k \|T^{(k)} - \bar{T}\|^2$ *with* $\bar{T} = K^{-1}\sum_k T^{(k)}$*. Then* $C = 0$ *if and only if all trajectories are identical. Consequently* $\min C = 0$ *is attained by collapse, and* $\partial C / \partial T^{(k)}$ *points toward the consensus, so gradient descent on* C *has a zero-spread stationary point that is a global minimum.*

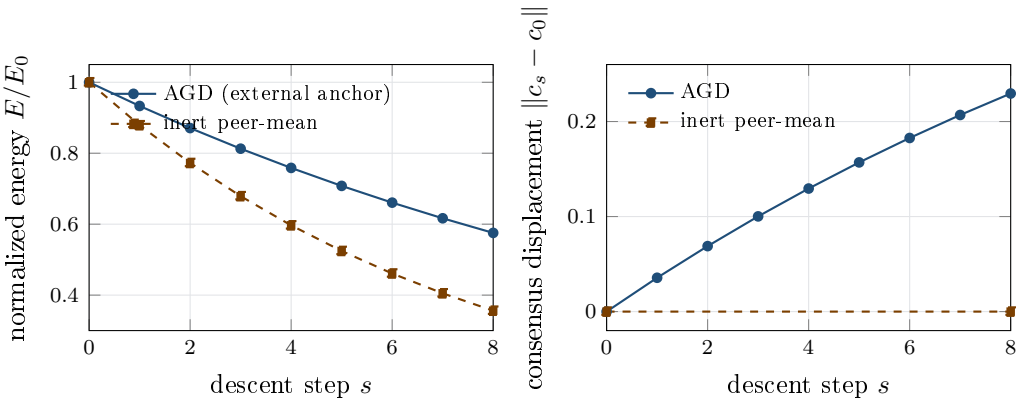


Figure 7: The central ablation, $S = 8$, $\eta = 0.5/S$, gates $g \in \{0.55, 0.60, 0.48\}$. **Left:** optimized energy decreases in both variants, and faster for the broken one—energy alone cannot detect the disconnection. **Right:** only the readout differs. AGD contracts the consensus toward the fixed anchor, displacing it by 0.2295; the inert peer-mean preserves the mean exactly, so its curve is identically zero and its logits are bit-identical to $S = 0$. This is why the ablation is reported as a retained contrast rather than removed: it is the cheapest demonstration that a refinement mechanism must be connected to the output to be measurable at all.

Proof. $\|\cdot\|^2 \geq 0$ with equality iff the argument is 0; hence C is a sum of non-negative terms and vanishes exactly when every $T^{(k)} = \bar{T}$, which forces all $T^{(k)}$ equal. For differentiability, $\frac{\partial C}{\partial T^{(k)}} = \frac{2}{K}(T^{(k)} - \bar{T}) - \frac{2}{K^2}\sum_j (T^{(j)} - \bar{T}) = \frac{2}{K}(T^{(k)} - \bar{T})$, which is 0 precisely at $T^{(k)} = \bar{T}$ and points inward toward it otherwise. □

This is the transfer of the RL collapse mechanism (Section 2.5) into our loss: the penalty is a collapse detector that the optimizer is rewarded for defeating. The fix (22) inverts the direction of the constraint, from "penalize disagreement" to "penalize the *absence* of disagreement", which has its minimum at a healthy spread.

Proposition 4 (The floor is non-degenerate)**.** *For $\rho > 0$ the term $g(\rho) = [\max(0, \rho_{\min} - \rho)]^2$ satisfies $g(\rho) = 0$ for all $\rho \geq \rho_{\min}$, is strictly decreasing on $(0, \rho_{\min}]$, and $g(\rho) \to \rho_{\min}^2 > 0$ as $\rho \to 0$. Hence its global minimum is attained on the* non-collapsed *set $\{\rho \geq \rho_{\min}\}$, not at collapse.*

Proof. Immediate: for $\rho \geq \rho_{\min}$ the inner maximum is 0, so $g \equiv 0$; on $(0, \rho_{\min}]$ the inner term is $\rho_{\min} - \rho \geq 0$, strictly decreasing in ρ, so g is strictly decreasing there with supremum $\rho_{\min}^2$ at $\rho = 0^+$. □

One honest caveat: the floor constrains only the *lower* bound on spread. It does not push trajectories apart when $\rho \geq \rho_{\min}$, so it is a collapse guard rather than a diversity objective. This is intentional, since an active diversity term is exactly the term that risks trading correctness for disagreement. It is also why $\rho_{\min}$ appears in the ablation grid.

4.3 Defect 3: the evaluation plan was not measurable at this scale

RQ2 and RQ3 were originally specified on GSM8K and HumanEval. At 12–46M parameters these tasks sit at chance: the smallest model reported to exceed 80% on GSM8K is 34B, and 1.3B plus a trained verifier was needed to reach 81.5% [59]. A pre-registered comparison on a floor-performing benchmark cannot decide anything; it can only return noise that will be over-interpreted. The fix is a strict separation between a *runnable* decision set (perplexity, ARC-easy, HellaSwag, LAMBADA, and synthetic known-answer multi-step items) and a *scaling-held-out* set (GSM8K, HumanEval, MMLU) that is reported but never used as a gate (Section 7).

4.4 Defect 4: an ambiguous discretization convention

Our own worked example cited the specification as using exact ZOH while in fact computing the first-order form, an inconsistency worth 43% on that example. The resolution and the error analysis are in Section 3.2.1. The general lesson, applied to the rest of this document, is that a specification and an implementation must agree on which convention is in force before any number derived from either is quoted.

5 Data

The corpus pipeline is a component of this thesis because reproducibility at this scale depends on data more than on architecture.

Registry and mixture. 54 primary dataset entries (plus 4 fallbacks) in 8 categories whose target weights sum to 1.0: code 30%, web text 20%, documentation 15%, wiki 10%, math 10%, science 5%, books 5%, structured knowledge 5%. Documentation is scraped from 18 sources with per-source timeouts and a FineWeb-Edu fallback chain.

Eight pipeline stages. (1) registry construction; (2) streaming with fallbacks; (3) quality, deduplication and contamination filtering; (4) abstract-syntax-tree filtering; (5) boilerplate stripping; (6) weighted sampling to hit the mixture; (7) sequence packing; (8) tokenization and caching.

Packing. Sequences are packed to 2048 tokens with fewer than 5% padding, achieved by segment-aware packing rather than naive concatenation.

Tokenizer. A 32,768-entry byte-pair-encoding vocabulary is trained in-house [63] on a registry sample deliberately skewed toward code to match the 0.30 code weight, and the *same* tokenizer is used for all three sizes so that comparisons are not confounded by tokenization.

6 Training

All three sizes share one recipe, so that size is the only variable.

Setting	Value
Steps	50,000
Tokens	$\approx$ 1.64B (32,768 tokens/step: batch 16×2048)
Learning rate	10^{-4}, cosine decay [61]
Warmup	3,000 steps
Effective batch	16 sequences (4×4 accumulation, 1 GPU)
Optimizer	AdamW, decoupled weight decay 0.1 [60]
Gradient clipping	1.0
Precision	BF16 parameters; scan state and Δ accumulation in FP32
Sequence length	2048
Hardware	1$\times$ A100, no distributed strategy

The token budget is deliberately *above* the compute-optimal ratio for these sizes [62]: at 12–46M parameters the model is in the over-trained regime, because the questions we ask (*does the mechanism do anything at all?*) are not compute-limited questions, and a badly under-trained model would confound "the mechanism does not work" with "the model has not learned the task yet". This choice is reported as a deviation from Hoffmann et al. [62] rather than hidden.

Stages proceed pretrain → SFT → DPO-lite [65]. Checkpoints are written per stage, and a smoke test (50 steps, loss-drop check, save/load/resume, $O(1)$ shape assertions) must pass before any full run.

7 Evaluation protocol

7.1 Principle: cost-matched, and pre-registered

Every comparison in this thesis is fixed before the run. Three comparisons are mandatory for any sandbox claim, and reporting only the first is considered a failed evaluation:

1. sandbox versus greedy decoding;
2. sandbox versus self-consistency majority vote [31] at equal FLOPs and equal token budget;
3. **sandbox versus a depth-matched model with no sandbox**, at the same parameter count and the same block-pass budget.

The third control exists because of Section 8: when compute is moved into extra layers rather than into a refinement mechanism, depth often wins, and without this control the two are indistinguishable.

7.2 Validation harness

An eight-phase harness gates production readiness before any full pretraining: (1) dataset registry verification, expecting $> 90\%$ available weight; (2) documentation build across 18 sources; (3) documentation quality (duplicate rate $< 5\%$, short-document rate $< 10\%$); (4) token distribution within $\pm 3\%$ of target per category; (5) decoded-sample inspection for malformed UTF-8, license spam, excessive newlines, repetition, and dirty padding; (6) packing quality with $< 5\%$ padding; (7) fallback-chain verification; (8) checkpoint save/load/resume smoke test. The harness returns one of READY, READY WITH MINOR WARNINGS, or NOT READY.

7.3 RQ1: memory

1. **Bounded-state proof.** Forward passes at $T = 512/2048/4096$ must produce identical `compressed` shapes $[B, 3d]$ and flat peak memory ($\pm 5\%$), with tokens per second reported against the exact-attention baseline at each length [19].
2. **Mechanism checks.** Selective-copying accuracy and induction-style extrapolation, where selective dynamics must clearly beat static gating—if they do not, L1's core premise fails.
3. **Backbone ablation.** L1 versus a small Transformer at equal parameters and equal tokens: perplexity, throughput, memory versus length.
4. **Recall probe.** One exact-recall task where attention is expected to win, reported as the bounded limitation.

7.4 RQ2: reasoning

Benchmarks are split by whether they can discriminate at this scale (Section 4.3).

Role	Tasks	Use
Runnable (XS/S/M)	validation and test perplexity; ARC-easy; HellaSwag; LAMBADA; synthetic known-answer multi-step items	the *decision gate*
Scaling-held-out	GSM8K; HumanEval; MMLU	reported, never the gate

Ablations, each mapped to a knob: $K \in \{1, 4, 8\}$ (with K=1 the chain-of-thought control); $R \in \{0, 2, 3\}$; $S \in \{0, 4, 8\}$ (with S=0 the inert-mechanism control of Section 4.1); diverse versus copy initialization; mean versus lowest-anchor-energy consensus; $\rho_{\min} \in \{0.01, 0.05, 0.2\}$; and **anchor versus peer-mean** target, which isolates the correction of Defect 1.

7.5 RQ3: honesty

1. **Calibration.** Agreement $\hat{a} = 1/(1 + V)$ plotted against per-example accuracy, with expected calibration error reported. This is the measurement that decides whether consensus is a usable confidence signal.

2. **Collapse monitoring.** ρ reported as a learning curve. A falling ρ is the failure signal and must be visible in the training logs, not reconstructed afterwards.

3. **Overturn accounting.** Per debate round, counts of wrong→right versus right→wrong transitions. Majority pressure suppressing corrections appears here first [34].

4. **Reranking.** Step-energy rerank versus final-vote, plus rationale-aware early stopping [46] for the efficiency table.

7.6 Results

All tables below are placeholders. They are included so the required content and its provenance are fixed in advance; no number is reported until the run produces it.

Table	Status
Parameter split per size	*[PENDING]* — confirm with `estimate_model_size`
Peak memory at $T = 512/2048/4096$	*[PENDING]* — bounded-state proof
Tokens/sec versus exact attention	*[PENDING]*
Cost-matched greedy / self-consistency / sandbox / depth-matched	*[PENDING]*
K, R, S ablations	*[PENDING]*
$\rho_{\min}$ sweep and anchor versus peer-mean	*[PENDING]*
Agreement–accuracy calibration (RQ3)	*[PENDING]*
Layer ladder: backbone-only → +L1 → +L1+L2	*[PENDING]*

8 Prior art, and the strongest evidence against us

The novelty audit's most important finding is not a supporting citation. Rauba, Fanconi and van der Schaar [52] run a compute-matched ladder from a plain Transformer to an autoregressive recursive model, holding block design, token stream, objective, and total block-pass budget fixed,

and report that the full recursive variant shows *no consistent benefit*. What generalizes best is untied depth and a *flat two-stream recurrent baseline*.
Three consequences, stated plainly:

1. **Stage L2 cannot be justified by appeal to that line of work.** Its own conclusion cautions against investing in the mechanism. Any argument that leans on it as evidence that latent refinement helps would be using the paper against its finding.

2. **The encouraging signal is structural, not mechanistic.** What worked there was separating streams and reading out with a weighted combination, which is closer to L2's shape than to nested halting—and is independently corroborated at comparable scale by the dual-stream result of Kerce et al. [51]. Two independent sources therefore support *stream separation*, which is the part of stage L2 we defend. The *recursive refinement* part is the hypothesis under test.

3. **The bar for our ablation is higher than we originally set.** The informative control is depth-matched-no-sandbox, not sandbox-versus-greedy (Section 7).

The expected result is correspondingly weaker than it was before this audit: the honest prior from the best-matched study in our own library is *no reliable gain*. RQ2 is therefore written as a falsifiable question with a genuine chance of a negative answer, and the thesis pre-commits to reporting a tie.

9 Limitations and threats to validity

Scale. At 12–46M parameters, absolute benchmark numbers will be far below published frontier results. The claims are about *mechanisms* moving a metric, never about state-of-the-art quality.

Recall. A fixed-size recurrent state is a lossy summary. Exact associative recall is expected to lose to attention; we report the recall probe rather than omit it [18].

Collapse and false consensus. K trajectories agreeing can be jointly wrong. ρ detects *collapse*, not *error*; a diverse but uniformly wrong set will not trip the floor. This is why RQ3 measures calibration rather than assuming agreement implies correctness.

Compute cost. Stage L2 costs $\times(2.6)$ refinement passes over the backbone at default settings (23). Gains must be reported against self-consistency at equal budget, not against greedy.

Single configuration. One recipe, one seed (42) in the reported protocol. Run-to-run variance is not characterized; differences smaller than seed variance should not be interpreted.

No kernels. Plain PyTorch means the selective scan is memory-bandwidth-bound and leaves throughput on the table. This is a deliberate auditability trade, not an oversight.

Deviation from compute-optimal scaling. We train well past the Chinchilla ratio [62] deliberately (Section 6).

10 Reproducibility

Seed 42; ≈ 13 configuration knobs, each mapped to an ablation; pinned dependencies; a warm cache for the corpus; pinned tokenizer per run; and a claim-to-evidence index mapping every load-bearing statement to either a locally stored paper or a hand derivation. The accompanying

markdown ("`thesis.md`") retains clickable links to the full vault, the paper library, and the per-claim proof index.

This document is self-contained LaTeX and compiles with a stock TinyTeX (TeX Live 2026) installation with no additional packages, via `pdflatex thesis.tex` run three times (once for the table of contents, once to settle cross-references, once to confirm convergence). The build is warning-clean: no errors, no undefined references or citations, and no overfull boxes.

11 Conclusion

Ekdant separates three concerns that are usually entangled: bounded memory, auditable computation, and honest confidence. Stage L1 replaces the unbounded per-token cache with a fixed-size selective state; stage T restores bounded local exact retrieval; stage L2 makes reasoning inspectable by materializing several live hypotheses and lets them confront one another.
The substantive result of this thesis is as much about the corrections as about the architecture. A peer-mean energy loop that provably cannot change the model's output, and an agreement loss that is minimized by the collapse it is meant to detect, were both in the original design and both would have produced publishable-looking negative results. Replacing them with an externally anchored, gated descent and a dimensionless collapse floor yields a mechanism that provably decreases energy, verifiably moves the readout, and cannot be satisfied by collapse. An audit that found the strongest published evidence against its own hypothesis—and pre-committed to reporting a tie—is, we would argue, the minimum standard for a claim about a new mechanism.

References

[1] A. Gu et al. How to train your HiPPO: state space models as implicit recurrent regularization. NeurIPS 2020. arXiv:2008.07669.

[2] J. Tay et al. Efficiently learning long sequences with structured state spaces. ICLR 2021. arXiv: 2011.04006.

[3] A. Gu et al. Efficiently modeling long-range dependencies with structured state spaces. ICLR 2022. arXiv:2111.00396.

[4] A. Gu et al. On the parameterization and initialization of diagonal state space models. NeurIPS 2022. arXiv:2206.11893.

[5] A. Gupta et al. DSS: efficiently learning with diagonal state space models. 2022. arXiv:2203.14343.

[6] S. Smith et al. Simplified state space layers for sequence modeling. ICLR 2023. arXiv:2208.04933.

[7] A. Gu and T. Dao. Mamba: linear-time sequence modeling with selective state spaces. 2023. arXiv: 2312.00752.

[8] T. Dao and A. Gu. Transformers are SSMs: generalized models and efficient algorithms through structured state space duality. ICML 2024. arXiv:2405.21060.

[9] D. Nguyen et al. S4ND: modeling long sequences with deep state space models. NeurIPS 2022. arXiv:2205.13493.

[10] R. Hasani et al. Liquid time-constant networks. AAAI 2023. arXiv:2209.12951.

[11] M. Poli et al. Hyena hierarchy: towards larger convolutional language models. ICML 2023. arXiv: 2302.10866.

[12] D. Fu et al. H3: hungry hungry hippos: towards language modeling with state space models. ICLR 2023. arXiv:2208.01814.

[13] Y. Sun et al. Retentive network: a successor to transformer for long-context general sequence modeling. 2023. arXiv:2307.08621.

[14] B. Peng et al. RWKV: reinventing RNNs for the transformer era. EMNLP 2023. arXiv:2305.13048.

[15] S. De et al. Griffin: mixing gated linear recurrences with local attention. ICML 2024. arXiv: 2402.19427.

[16] B. Beck et al. xLSTM: xLSTM: a recurrent language model. NeurIPS 2024. arXiv:2405.04517.

[17] J. Wang et al. MambaByte: tokenizing is all you need. 2024. arXiv:2401.13660.

[18] O. Lieber et al. Jamba: a hybrid Transformer-Mamba language model. 2024. arXiv:2403.19887.

[19] T. Dao et al. FlashAttention: fast and memory-efficient exact attention with IO-awareness. NeurIPS 2022. arXiv:2205.14135.

[20] T. Dao. FlashAttention-2: faster attention with better parallelism and work partitioning. 2023. arXiv:2307.7212.

[21] A. Vaswani et al. Attention is all you need. NeurIPS 2017. arXiv:1706.03762.

[22] J. Su et al. RoFormer: enhanced transformer with rotary position embedding. 2021. arXiv: 2104.09864.

[23] N. Shazeer. Fast transformer decoding: one write-head is all you need. 2019. arXiv:1911.02150.

[24] J. Ainslie et al. GQA: training generalized multi-query transformer models from multi-head checkpoints. EMNLP 2023. arXiv:2305.13245.

[25] N. Shazeer. GLU variants improve transformer. 2020. arXiv:2002.05202.

[26] B. Zhang and R. Sennrich. Understanding deep learning requires rethinking normalization. 2019. arXiv:1910.07467.

[27] R. Xiong et al. On layer normalization in the transformer architecture. ICML 2020. arXiv: 2002.04745.

[28] A. Jiang et al. Mistral 7B. 2023. arXiv:2310.06825.

[29] B. Peng et al. YaRN: efficient context window extension of large language models. 2023. arXiv: 2309.00071.

[30] J. Wei et al. Chain-of-thought prompting elicits reasoning in large language models. NeurIPS 2022. arXiv:2201.11903.

[31] X. Wang et al. Self-consistency improves chain of thought reasoning in language models. ICLR 2023. arXiv:2203.11171.

[32] Y. Du et al. Improving factuality and reasoning in language models through multiagent debate. 2023. arXiv:2305.14325.

[33] P. Liang et al. Encouraging divergent thinking in large language models through multi-agent debate. EMNLP 2023. arXiv:2305.19118.

[34] W. Wu et al. Can agents really debate? 2025. arXiv:2511.07784.

[35] S. Yao et al. Tree of thoughts: deliberate problem solving with large language models. NeurIPS 2023. arXiv:2305.10601.

[36] M. Besta et al. Graph of Thoughts: solving elaborate problems with large language models. AAAI 2024. arXiv:2308.09687.

[37] D. Zhou et al. Least-to-most prompting techniques to enable complex reasoning in large language models. ICLR 2023. arXiv:2205.10625.

[38] S. Yao et al. ReAct: synergizing reasoning and acting in language models. ICLR 2023. arXiv: 2210.03629.

[39] N. Shinn et al. Reflexion: language agents with verbal reinforcement learning. NeurIPS 2023. arXiv: 2303.11366.

[40] A. Madaan et al. Self-Refine: iterative refinement with self-feedback. NeurIPS 2023. arXiv: 2303.17651.

[41] H. Lightman et al. Let's verify step by step. 2023. arXiv:2305.20050.

[42] A. Uesato et al. Improving math reasoning with process-based feedback. 2022. arXiv:2211.14275.

[43] Z. Huang et al. Large language models cannot self-correct reasoning yet. ICLR 2023. arXiv: 2210.11610.

[44] E. Zelikman et al. STaR: bootstrapping reasoning with reasoning. NeurIPS 2022. arXiv:2203.14465.

[45] A. Chan et al. ChatEval: a platform for evaluating LLMs as judges. 2023. arXiv:2308.07201.

[46] G. Wan et al. Reasoning-aware self-consistency. NAACL 2025.

[47] A. Smit et al. Should we be going MAD? A look at agentic multi-agent debate. ICML 2024. PMLR 235.

[48] Energy calibration for implicit chain of thought. 2025. arXiv:2511.07124.

[49] S. Burtsev and A. Rumshisky. Multi-stream transformers: sequential knowledge sharing. 2021. arXiv:2107.10342.

[50] N. Shazeer. Talking-heads attention. 2020. arXiv:2003.02436.

[51] J. Kerce and E. Fox. Dual-stream transformer. 2026. arXiv:2603.07461.

[52] P. Rauba, C. Fanconi and M. van der Schaar. Tiny autoregressive recursive models. 2026. arXiv: 2603.08082.

[53] J. Jolicoeur-Martineau. Less is more: recursive reasoning with tiny networks. 2025. arXiv: 2510.04871.

[54] Energy transformer. NeurIPS 2023.

[55] NRGPT: an energy alternative for GPT. 2025. arXiv:2512.16762.

[56] W. Wang et al. Mixture-of-agents enhances large language model capabilities. ICLR 2025.

[57] Y. Li et al. On the mode collapse of KL-regularized reinforcement learning. 2026. arXiv:2510.20817.

[58] Entropy collapse in RL with verifiable rewards. ACL 2026 Findings 1266.

[59] Z. Li et al. TinyGSM: grading intermediate reasoning for small language models. 2023. arXiv: 2312.09241.

[60] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. ICLR 2019. arXiv:1711.05101.

[61] I. Loshchilov and F. Hutter. SGDR: stochastic gradient descent with warm restarts. ICLR 2017. arXiv:1608.03983.

[62] J. Hoffmann et al. Training compute-optimal large language models. 2022. arXiv:2203.15556.

[63] R. Sennrich et al. Neural machine translation of rare words with subword units. ACL 2016. arXiv: 1508.07909.

[64] C. Guo et al. On calibration of modern neural networks. ICML 2017. arXiv:1706.04599.

[65] D. Rafailov et al. Direct preference optimization. NeurIPS 2023. arXiv:2305.18290.